ETHNOHACK 2026-Track 2: Natural Products and Drug Discovery

University of Tunis El Manar (UTM)

POMENOVA

NATURE MEETS INTELLIGENCE



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Imagine someone you love struggling to recognize you

Alzheimer's disease progressively affects memory and cognitive function, disrupting communication, independence, and everyday life.

57M

People were living with dementia worldwide in 2021.

60–70%

Of dementia cases are attributable to Alzheimer's disease.

~10M

New dementia cases occur worldwide each year.

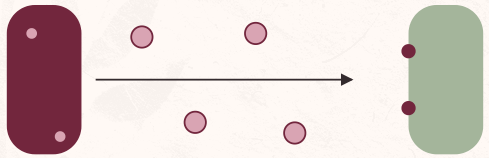
> Current treatments can help manage symptoms, but they do not provide a cure.

> Source:World Health Organization(WHO),Dementia Fact Sheet.



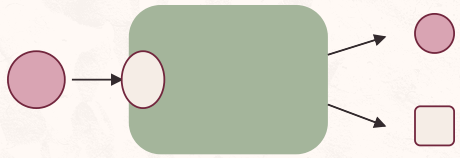
Why AChE and pomegranate?

Cholinergic signalling



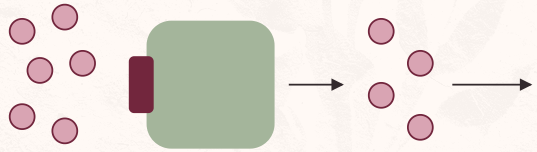
Acetylcholine (ACh) activates cholinergic receptors, supporting neuronal communication.

AChE hydrolysis



Acetylcholine + H_2O → Choline + Acetate
AChE terminates cholinergic signalling by hydrolyzing acetylcholine.

AChE inhibation



Inhibiting AChE reduces acetylcholine degradation and prolongs its availability at cholinergic synapses.

WHY TARGET AChE?

Acetylcholine supports neuronal communication; AChE hydrolyses it into choline and acetate. Reducing breakdown can support available cholinergic signalling.

Donepezil and galantamine are clinically used AChE inhibitors for symptomatic management of Alzheimer's disease.

Target: human AChE · ChEMBL CHEMBL220 · UniProt P22303

WHY POMEGRANATE?

Pomegranate (*Punica granatum L.*) is familiar in Tunisia. Morzelle et al. (2019) reported AChE inhibition by pomegranate peel phenolic extracts—plant-level investigation, not proof for every purified molecule.

Sources:*Morzelle et al., Journal of Food Bioactives (2019), DOI: 10.31665/JFB.2019.5181; PubChem/UniProt P22303.*

Research gap:
Extract-level activity does not establish the inhibitory potency, selectivity, or brain exposure of individual compounds.

Research question
Can machine learning prioritize ten selected pomegranate compounds for further investigation against human AChE?

DATA COLLECTION AND CURATION

From PubChem to ChEMBL: Building Our Molecular Datasets

Compound	PubChem CID	MW, g/mol	TPSA, Å²
Punicalagin	16129719	1084.7	526.6
Punicalin	5464368	782.5	385.2
Corilagin	73568	634.5	310.7
Pedunculagin	442688	784.5	377.4
Ellagic acid	5281855	302.2	141.3

Compound	PubChem CID	MW, g/mol	TPSA, Å²
Gallic acid	370	170.1	98.0
Catechin	9064	290.3	110.4
Epicatechin	72276	290.3	110.4
Quercetin	5280343	302.2	131.4
Rutin	5280805	610.5	269.4

INITIAL CHEMBL 37 DATA EXTRACTION

7,826 measurement rows

6,287 distinct molecules ·

600 molecules with repeated-measurement

146 molecules with a within-molecule pIC50 range > 1

PubChem: identities, CIDs, formulas, SMILES. ChEMBL 37: experimental human AChE IC50 measurements for training and evaluation. RDKit: descriptors and fingerprints from structures. PubChem identity does not prove occurrence or abundance in a particular Tunisian sample.

FINAL CURATED ML EXPORT:

4,859

molecules prepared for machine-learning training and evaluation.

The 10 selected compounds are prediction candidates, distinct from the 4,859-molecule training dataset. Computational properties and predictions do not constitute experimental evidence of AChE inhibition.

METHODOLOGY

From Molecular Structures to AI Predictions

A reproducible six-step computational workflow.

01

Curate

Inspect missing values, duplicates, repeats and stereochemistry.

02

Transform

Convert experimental human AChE IC50 values from nM into pIC50.
Display the exact formula:
pIC50 = 9 − log10(IC50 in nM)

03

Encode

Generate chirality-aware Morgan fingerprints (radius 2, 2,048 bits). Calculate molecular descriptors with RDKit.

04

TRAIN

Compare Random Forest and XGBoost regression models using the curated ChEMBL dataset.

05

Evaluate

Evaluate held-out predictions using an initial canonical-structure GroupShuffleSplit and a separate Bemis–Murcko scaffold split. Report MAE and R^2 on the following results slides.

06

Prioritize

Apply the selected model to ten pomegranate-associated compounds and examine predicted pIC50 and training-set similarity.

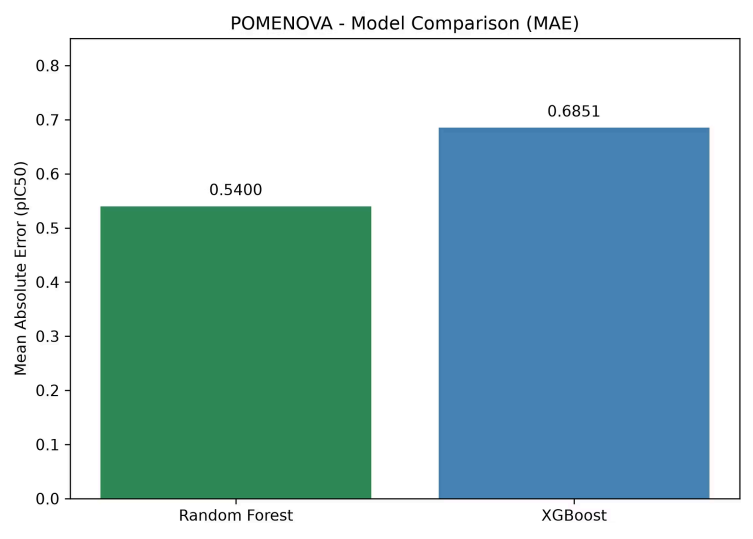
"Experimental IC50 values train the model; candidate pIC50 values are computational predictions, not experimentally confirmed inhibition."

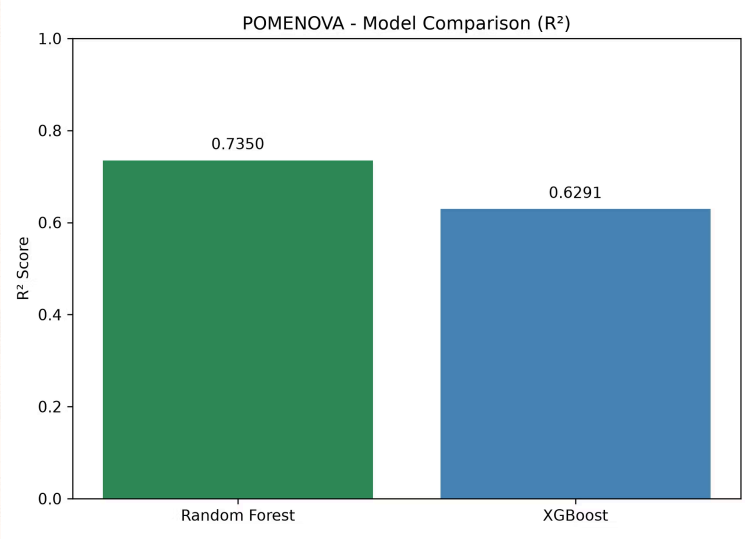
MODELMODEL COMPARISON

Two Models.One Clear Leader

"Initial grouped holdout validation-predicting human AChE pIC50"

i





MAE, pIC50 units · lower is better

R^2 · higher is better relative to the held-out mean baseline

Model	MAE	R^2
Random Forest	0.5400	0.7350
XGBoost	0.6851	0.6291

Selection is conditional on this reported initial grouped holdout, not universal superiority or statistical significance.

"21.2% lower MAE than XGBoost"
"+0.1059 higher R^2"

Scientific limitation:
"Model selection is conditional on this evaluation. Independent scaffold-based validation is reported separately; these results do not establish statistical significance."

MODEL ROBUSTNESS

"Beyond Familiar Molecules: Testing AI Generalization"

-Bemis–Murcko scaffold-based holdout validation-

"Can our model predict molecules with different chemical frameworks? Scaffold splitting separates molecular core structures between training and testing, reducing structural overlap."

SCAFFOLD EVALUATION

4,859 molecules
2,103 scaffold groups
3,901 training molecules
958 testing molecules

Random Forest ·

scaffold MAE =0.5935

R^2 =0.6777

XGBoost ·

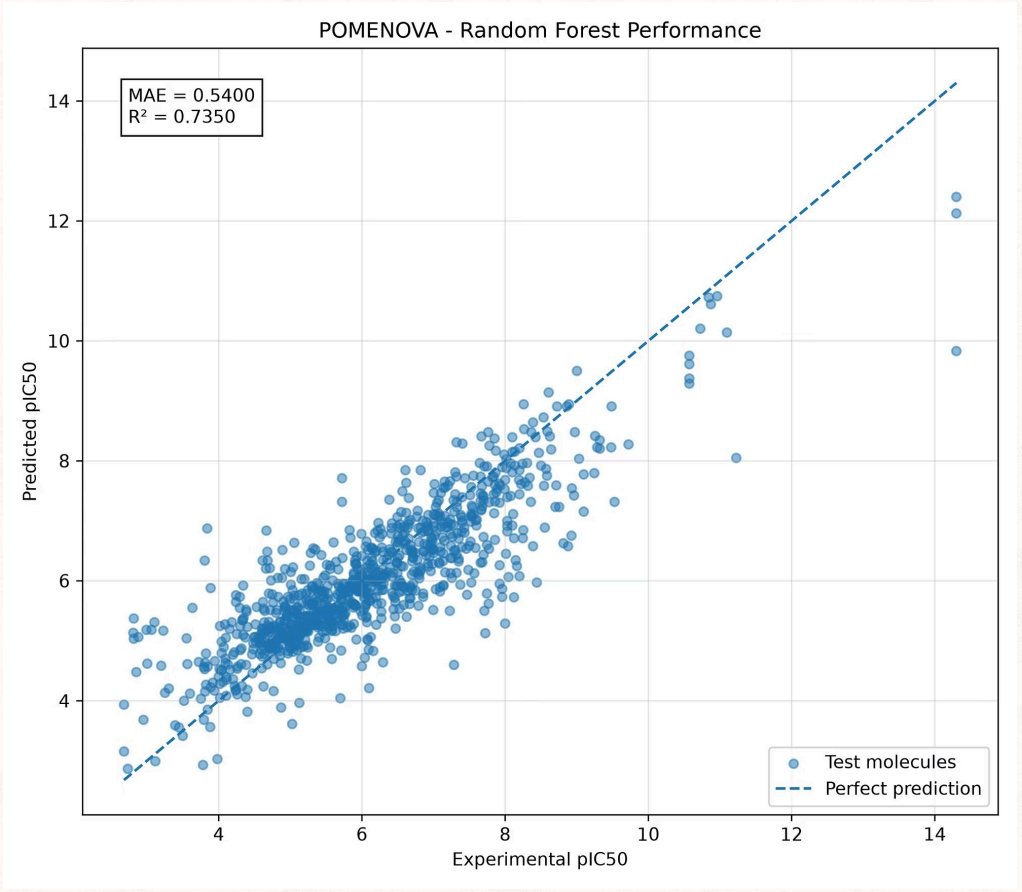
scaffold MAE =0.7013

R^2 =0.5924

> **"Random Forest maintained better predictive performance on the scaffold holdout."**

> **"Single scaffold split; further validation across seeds, independent datasets and experimental assays is required."**

Initial grouped holdout



Random Forest performance on the initial split

CANDIDATE PREDICTIONS

“From 10 Natural Compounds to Prioritized Research Candidates”

-AI-predicted AChE inhibitory potency, molecular similarity, and uncertainty-

Quercetin:

6,886 : Predicted pIC50

≈ 130 nM: Predicted IC50

Max Tanimoto: 0.718

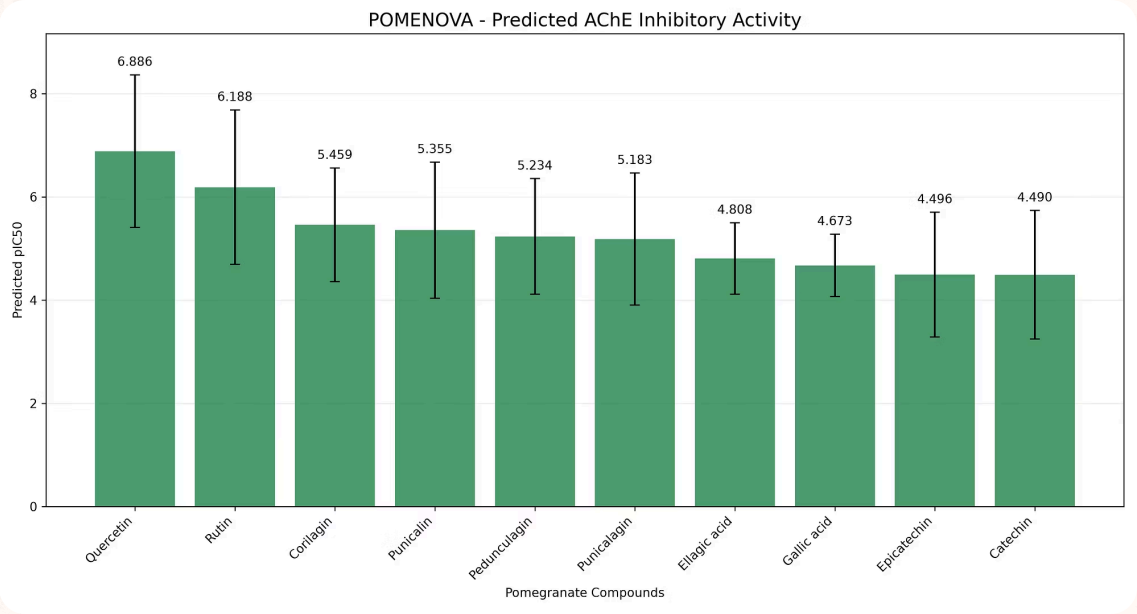
Rutin:

6,188 : Predicted pIC50

≈ 648 nM: Predicted IC50

Max Tanimoto: 0.333

The model ranks quercetin highest among the ten candidates. However, seven compounds have maximum training-set Tanimoto similarity below the team's exploratory threshold of 0.4, indicating that many predictions involve relatively unfamiliar molecular structures. These findings guide experimental prioritization rather than establish biological activity.



Predictions do not establish activity, brain penetration, safety or clinical benefit.

Higher predicted pIC50 corresponds to lower predicted IC50. Error bars show dispersion between Random Forest trees, not calibrated confidence intervals

POMENOVA APPLICATION

Explore, predict and explain



EXPLORE

Browse 10 compounds and inspect their structures and properties.

PREDICT

View predicted AChE potency and similarity to training molecules.

EXPLAIN

Review model performance, supporting evidence and limitations.

Computational research prototype. Predictions require experimental validation.

WHAT COMES NEXT?

“One Better Question Can Inspire the Next Discovery.”

POMENOVA combines Tunisian botanical heritage, molecular chemistry and machine learning to prioritize natural compounds for future investigation against human acetylcholinesterase.

“Our contribution is not a cure. It is a reproducible starting point for better-informed research.”

> “We are not claiming that a pomegranate can cure Alzheimer's. We are showing how nature and intelligence can help us ask better scientific questions.”

LIMITS

Assay heterogeneity · Limited external validation · Structural extrapolation · No experimental confirmation · Unknown safety and brain exposure.



01 — Strengthen the evidence:

Audit literature and compound identities, repeat validation across random seeds, and evaluate external chemical datasets.

02 — Validate experimentally:

Test selected compounds using human AChE dose–response assays with appropriate interference controls.

03 — Investigate therapeutic relevance:

Assess selectivity, toxicity, pharmacokinetics and potential central nervous system exposure.